Explainable Machine Learning for Student Burnout Risk Stratification and Predictive Screening: A Mixed-Methods Study with Qualitative Triangulation

Abstract

Background: Academic burnout impairs cognitive performance, student retention, and psychological well-being in tertiary education [1–4]. Institutional responses remain largely reactive, while machine learning screening tools frequently operate as uninterpretable black-boxes [43, 63].

Methods: We deployed an explanatory sequential mixed-methods design (N = 601 survey; N = 20 interviews) across 11 universities in Bangladesh [10, 11, 61, 19, 20]. Grounded in Conservation of Resources (COR) [13, 14] and Job Demands–Resources (JD-R) [15, 16] frameworks, we engineered nine composite features capturing circadian disruption, evaluative load, and coping reserves. Ten classifiers were benchmarked under nested 10-fold cross-validation with threshold calibration [36–42, 51]. TreeSHAP/KernelSHAP/LIME attributions [43–47] were triangulated against reflexive thematic analysis ($\kappa = 0.82$, 25% audit) [55–60].

Results: Severe burnout affected 42.43% (n = 255) of students. Random Forest was the best-performing model (65.89% accuracy, ROC-AUC = 0.7126, indicating moderate discriminative ability per Hosmer and Lemeshow [76]), outperforming the 57.57% majority baseline (McNemar $\chi^2 = 13.96$, $p < .001$) [54] and generalizing across degree tiers (holdout accuracy = 67.03%, AUC = 0.6597) [42]. Calibrating threshold to $\theta = 0.38$ elevated sensitivity to 71.76% (183/255 at-risk students identified). Global SHAP rankings identified Academic Performance Index (mean |SHAP| = 0.0388), CGPA bracket (0.0362), Screen-to-Sleep Ratio (0.0298), and Burnout Vulnerability Index (0.0238) as top drivers; sociodemographics contributed negligibly (|SHAP| < 0.007). Thematic analysis revealed four dimensions: Academic Dread and Career Despair, Institutional Identity and Low-Pressure Burnout, Digital Fatigue as Maladaptive Coping, and Circadian Collapse.

Conclusion: Student burnout is governed by dynamic behavioral imbalances—specifically nocturnal screen use displacing restorative sleep—and evaluative pressure rather than static demographics. We open-source our pipeline and deploy an interactive screening platform (*Burnout Radar*) for privacy-preserving institutional triage.

Keywords: Academic Burnout, Explainable Artificial Intelligence (XAI), SHAP, Mixed-Methods Research, Machine Learning, Higher Education.

1. Introduction

Academic burnout—comprising emotional exhaustion, cynicism, and reduced efficacy—is an escalating crisis in tertiary education [1–5, 68]. Digitization, credential inflation, and socioeconomic pressures intensify undergraduate distress [6–8]. In low- and middle-income countries (LMICs) such as Bangladesh, institutional session jams, rigid pedagogy, and socio-familial expectations compound vulnerability [9–11, 61].

Our work is based on the Job Demands–Resources (JD-R) [15–17] and Conservation of Resources (COR) [13, 14] frameworks. The JD-R model predicts that strain occurs when evaluative demands

(heavy coursework, examinations) outpace personal resources (restorative sleep, social support) [15, 72]. COR views burnout as a loss spiral that builds momentum with time: sacrificing sleep for academic deadlines depletes cognitive capacity, exacerbating anxiety and systemic exhaustion [13, 14, 22–25, 29].

Prior studies have significant limitations. Linear statistical analyses are unable to model non-linear behavioural thresholds such as the effects of screen-time on circadian disruption [26–28, 66, 67]. Conversely, educational ML studies often use opaque black-box models with no actionable clinical explanations [31–35, 43, 62, 63], and are removed from students' lived subjective experiences [19–21, 64, 65].

To address these gaps, we used an explanatory sequential mixed-methods design (QUAN → QUAL) across 11 universities in Bangladesh [19, 20], combining supervised ML, model-agnostic SHAP explainability [44, 45], and reflexive thematic analysis [56–58]. Contributions include: (1) mixed-methods architecture (N = 601 survey; N = 20 interviews) with structured triangulation [21]; (2) theory-driven COR/JD-R composite features benchmarked across 10 classifiers [36–42]; (3) model-agnostic SHAP attributions validated across degree tiers [42, 44, 45, 51]; and (4) open-access deployment of *Burnout Radar* for institutional screening.

2. Literature Review

2.1 Multidimensional Predictors of Academic Burnout

Academic burnout consists of emotional exhaustion, depersonalization/cynicism, and academic inefficacy [1–4]. The major determinants are heavy course loads, frequent assessments, and competitive grading [5, 66, 69]. Cumulative Grade Point Average (CGPA) governs scholarship eligibility, parental expectations, and employment prospects [5, 61]. Sleep restriction (<6 h/night) impairs prefrontal executive function [22–25], while nocturnal screen engagement triggers circadian phase-shifting and sleep displacement [26–28]. Demographics (gender, age, discipline) show inconsistent associations across contexts [29, 30, 70, 71].

2.2 Machine Learning and Explainable AI in Educational Mental Health

Supervised models (Random Forest, SVM, Gradient Boosting) predict student attrition, depression, and academic failure [31–35, 62]. However, adoption is impeded by the black-box dilemma [43]. SHAP provides mathematically consistent marginal feature attributions grounded in cooperative game theory [44–46, 48]. While established in healthcare, SHAP is rarely applied to educational screening with cross-validation stability checks [63].

2.3 Methodological Gaps and Rationale for Mixed-Methods Triangulation

As summarized in Table 1, prior work exhibits a divide: quantitative ML studies optimize predictive metrics without qualitative grounding [61–63], whereas qualitative studies offer rich narratives without scalable screening capacity [64, 65]. We bridge this dichotomy via an explanatory sequential mixed-methods design (QUAN → QUAL) [19, 20].

Table 1. Structured Methodological Comparison Against Prior Literature

Study	Target Focus	Sample (N)	Models Evaluated	Interpretability / XAI	Validation Strategy	Qualitative Strand	Triangulation Protocol
Mamun et al. (2019) [61]	Problematic Internet Use & Student Distress	2,032	Logistic Regression, Chi-square	Odds Ratio (AOR)	Not applicable	No	No
Gil et al. (2022) [75]	College Student Depression Risk & Predictive Factors	171 triads (N = 513)	Sparse Logistic Regression, SVM, Random Forest	Feature importance ranking	70/30 Train / Test Split	No	No
Iatrellis et al. (2021) [34]	Academic Performance & Dropout	Institutional cohort	K-Means + SVM, DT, ANN, RF, NB	Model weights / ranking	Train / Test Split	No	No
Islam et al. (2020) [10]	University Student Depression & Anxiety	3,122	Multivariate Logistic Regression	Adjusted Odds Ratios (AOR)	Not applicable	No	No
Hossen et al. (2023) [30]	University Student Psychological Distress	1,200	Binary Logistic Regression	Adjusted Odds Ratios (AOR)	Not applicable	No	No
Present study	University Student Academic Burnout	601	10 models + Soft Voting Ensemble	SHAP (Global Attribution & Local Risk Profiling) + PI + FI	10-Fold Stratified CV + Subgroup	Yes (N = 20 interviews)	Yes (Explanatory Matrix)

Note. LR = Logistic Regression; RF = Random Forest; SVM = Support Vector Machine; LDA = Linear Discriminant Analysis; KNN = K-Nearest Neighbors; NB = Naïve Bayes; DT = Decision Tree; ANN =

Artificial Neural Network; SHAP = SHapley Additive exPlanations; PI = Permutation Importance; FI = Feature Importance; CV = Cross-Validation.

3. Materials and Methods

3.1 Study Design and Research Framework

We implemented an explanatory sequential mixed-methods design (QUAN → QUAL) conforming to GRAMMS (Good Reporting of A Mixed Methods Study) guidelines and mixed-methods design standards [19, 20] across four phases (Figure 1): cross-sectional survey (N = 601), feature engineering and nested 10-fold ML benchmarking, SHAP interpretability with threshold calibration, and semi-structured qualitative interviews (N = 20) analyzed via reflexive thematic analysis [56–58]. Reporting checklists conforming to COREQ (Consolidated Criteria for Reporting Qualitative Research; qualitative strand; Supplementary S1) [52] and TRIPOD (prediction model; Supplementary S2) [51] reporting standards are provided as supplementary materials.

Figure 1: End-to-End Methodology & Machine Learning Architecture

Explanatory Sequential Mixed-Methods Architecture (QUAN → QUAL): Quantitative ML/XAI & Qualitative Triangulation

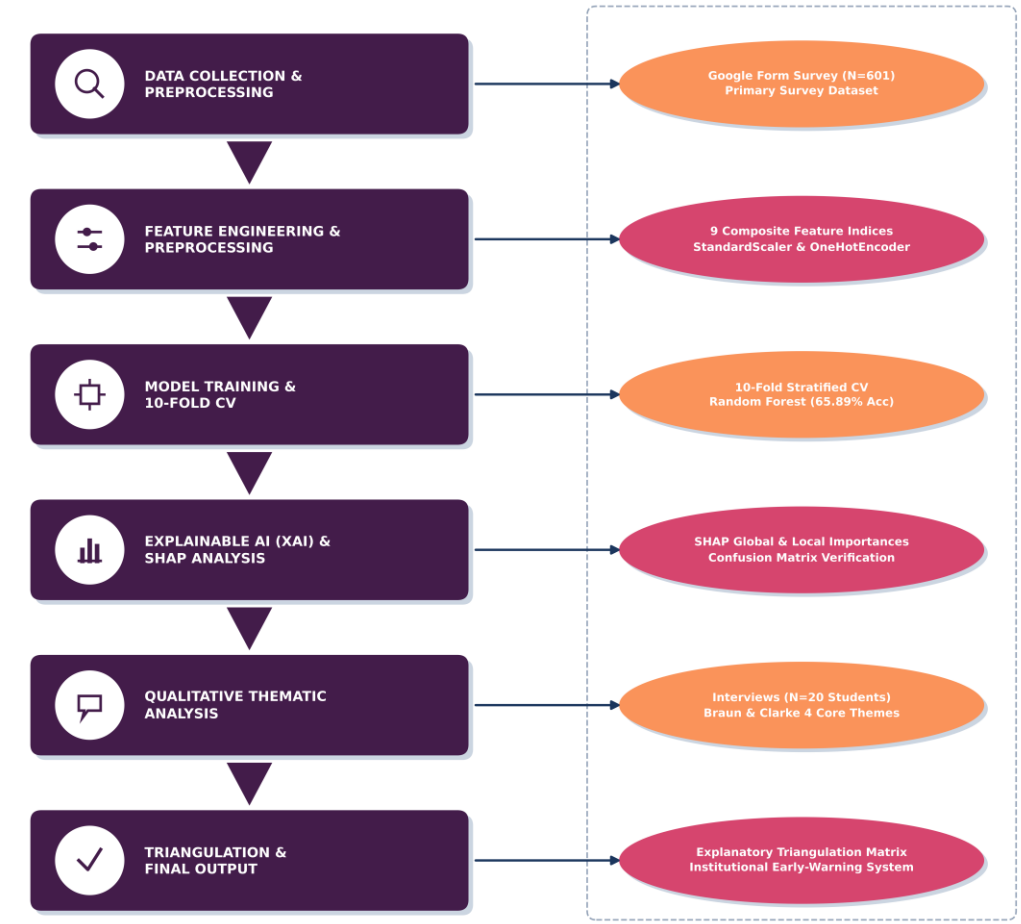


Figure 1: End-to-end mixed-methods research methodology workflow

Figure 1. Explanatory sequential mixed-methods research design workflow (QUAN → QUAL) integrating cross-sectional survey predictive modeling and semi-structured interview reflexive thematic analysis.

3.2 Quantitative Sample, Setting, and Ethics

The cohort comprises N = 601 tertiary students across 11 universities in Bangladesh (8 private, 1 public, 2 National University affiliated colleges) surveyed between October 2023 and February 2024. As an anonymous, non-interventional observational survey conducted with voluntary participation, this study satisfied educational research ethics exemption criteria under institutional guidelines (Department of Computer Science and Engineering, Presidency University, Dhaka, Bangladesh). No formal IRB approval number was assigned under the institutional exemption policy for non-clinical, anonymous educational research; the study adhered to the Declaration of Helsinki throughout. Participants provided electronic informed consent prior to data collection. The quantitative cohort employed convenience sampling through institutional email broadcasts, course announcements, and snowball referrals within each institution. Purposive maximum variation sampling was applied exclusively to the qualitative interview strand (N = 20) to ensure burnout-tier and degree-level diversity. The quantitative cohort may overrepresent technology-adjacent disciplines; generalizability to non-CSE majors and rural institutions warrants caution.

Sample Size and Power: With 255 high-burnout events across 18 raw survey features entering the primary regression-based analysis, the Events Per Variable (EPV) ratio was 255/18 = 14.17, exceeding the ≥ 10 EPV heuristic [36]. Note that the full ML feature space comprised 27 variables (18 raw + 9 engineered) and 37 dimensions after one-hot encoding of categorical variables; ML classifier performance under this expanded space was evaluated via stratified cross-validation, which does not require EPV compliance — EPV criteria are specific to logistic regression-type models and do not apply to tree-based ensemble classifiers evaluated via cross-validation [77]. Post-hoc power analysis (G*Power v3.1.9.7) confirmed that N = 601 achieves power (1 - β) 0.99 for small-to-moderate effects (w = 0.18, d = 0.35) at α = .05.

3.3 Variable Operationalisation and Target Criterion

The survey captured indicators across sociodemographic, academic, lifestyle, and psychological domains (Table 2).

Table 2. Survey Instrument Items, Operational Definitions, and Empirical Ranges (N = 601)

Variable Name	Domain / Category	Question / Item Operationalisation	Measurement Scale & Anchors	Empirical Range in Data
gender	Demographic	Self-reported gender identity (binary survey options: Male,	Categorical (Male, Female)	Male (57.24%), Female (42.76%)

		Female)		
age_group	Demographic	Student age bracket at survey completion	Categorical (17–18, 19–20, 21–22, 23–24, 25+ years)	17–18 to 25+ years
degree	Academic Profile	Current enrolled degree program	Categorical (Bachelor's, Master's, Ph.D., Diploma/Associate)	4 program tiers
academic_year	Academic Profile	Current academic year of study	Categorical (1st Year, 2nd Year, 3rd Year, Final Year)	4 study stages
study_hours_numeric	Behavioural	Average self-reported daily dedicated study duration	Binned / discrete midpoint (Hours per day)	0.50 – 7.00 hours
sleep_hours_numeric	Behavioural / Circadian	Average self-reported daily total sleep duration	Binned / discrete midpoint (Hours per day)	4.00 – 9.00 hours
social_media_hours	Behavioural / Digital	Average daily recreational screen time and social media use	Binned / discrete midpoint (Hours per day)	0.50 – 7.00 hours
physical_activity_hours	Lifestyle / Reserve	Average daily physical exercise, gym, sports, or active recreation duration	Binned / discrete midpoint (Hours per day)	1.00 – 5.00 hours
attendance_pct	Academic Engagement	Official self-reported institutional classroom attendance	Binned / discrete midpoint (%)	20.00% – 95.00%
cgpa_midpoint	Academic Performance	Cumulative Grade Point Average midpoint bracket	Binned / discrete midpoint (4.00 grading scale)	2.25 – 3.875 CGPA

stress_score	Psychological Demand	Perceived academic stress and mental pressure level	4-point Likert (1 = Low, 2 = Mild, 3 = Moderate, 4 = Severe)	1.00 – 4.00
depression_score	Psychological Demand	Self-reported depressive affect, sadness, and career despair	4-point Likert (1 = None/Minimal, 2 = Mild, 3 = Moderate, 4 = Severe)	1.00 – 4.00
academic_pressure_score	Academic Load	Perceived curriculum load, deadlines, and exam difficulty	4-point Likert (1 = Manageable, 2 = Moderate, 3 = High, 4 = Overwhelming)	1.00 – 4.00
workload_score	Academic Load	Perceived assignment volume and coursework density	4-point Likert (1 = Light, 2 = Moderate, 3 = Heavy, 4 = Extreme)	1.00 – 4.00
part_time_score	Employment Demand	Employment commitments outside academic coursework	3-point Ordinal (0 = None/Unemployed, 1 = Part-time, 2 = Substantial/Full-time)	0.00 – 2.00
motivation_score	Motivational Resource	Intrinsic motivation, academic drive, and course enthusiasm	3-point Ordinal (1 = Low/Disengaged, 2 = Moderate, 3 = High/Driven)	1.00 – 3.00
sleep_quality_score	Restorative Resource	Subjective sleep quality and restorative restfulness	3-point Ordinal (1 = Poor/Disturbed, 2 = Fair/Average, 3 = Good/Restful)	1.00 – 3.00
burnout_score	Target Variable	Global self-reported academic burnout severity tier	3-point Ordinal (1 = Low [n=117], 2 = Medium [n=229], 3 = High [n=255])	1.00 – 3.00 (Target: High=1, Low/Med=0)

Note. Empirical distributions across the quantitative cohort (N = 601). Interval midpoints were derived from standardized response brackets. Physical activity hours reflect ordinal brackets: < 1 hour/day (coded to 1.0 h/day to reflect light baseline locomotion/walking), 1–2 hours/day (coded 2.0 h), 2–4 hours/day (coded 3.0 h), and 4 hours/day (coded 5.0 h), capturing cumulative physical exertion rather than exclusive gym exercise.

Survey Design & Operationalisation: Standardized ordinal brackets were coded to interval midpoints. CGPA brackets were defined as: ≥ 2.00 to < 2.50 (midpoint = 2.25, assuming a minimum passing CGPA of 2.00), 2.50–3.00 (midpoint = 2.75), 3.00–3.50 (midpoint = 3.25), and 3.50 to 4.00 (midpoint = 3.75). Gender followed binary administrative classifications, with nuanced dimensions explored qualitatively (Section 8). Academic burnout was assessed via a 3-point screening metric adapted from the Maslach Burnout Inventory (MBI) [2, 49]: Mild (n = 117, 19.47%), Moderate (n = 229, 38.10%), and Severe (n = 255, 42.43%). For predictive triage, the criterion was operationalized as binary: *High Burnout* (Score 3; Target = 1, n = 255, 42.43%) versus *Low/Medium Burnout* (Scores 1 & 2; Target = 0, n = 346, 57.57%).

3.3.1 Psychometric Validity and Methodological Considerations

Psychometric Note: Single-item global burnout ratings offer high completion efficiency and acceptable concurrent validity (r = 0.64–0.76) [49, 50]. Although the present instrument captures global burnout severity rather than the MBI-SS three-factor structure (exhaustion, cynicism, academic efficacy) [4], single-item global ratings demonstrate acceptable criterion-related validity, consistent with the cited medical validation studies [49, 50]. It is important to note that both cited validation studies [49, 50] were conducted with medical professional (physician) samples rather than undergraduate student populations; psychometric properties may differ across populations, and student-specific validation evidence for single-item burnout tools remains more limited (cf. Dyrbye et al., 2008 [65]). We acknowledge this as a limitation (Section 9.3) and recommend future replication employing the validated MBI-SS [4]. Cross-sectional single-item measures remain subject to common-method variance (CMV) and an operational ceiling on ROC-AUC bounded by criterion reliability; comparisons against full multi-item inventories (e.g., MBI-SS, CBI) [4, 7] warrant caution.

Qualitative Guide & Discriminant Validity: A 10-item guide examined routines, sleep, screen use, employment, stress, mental health, exercise, burnout, attendance, and sleep quality (interviews: 20–30 min). Perceived depression correlated moderately with burnout ($r_s = 0.284$, $p < .001$), confirming 90% unshared variance and no collinearity (VIF = 2.04 < 3.0) [67, 68]. SHAP analysis confirmed that Academic Performance Index (|SHAP| = 0.0388) and Screen-to-Sleep Ratio (0.0298) far outweighed depression (0.0152), verifying domain-specific burnout measurement.

3.4 Feature Engineering

To capture complex behavioral interactions, we engineered nine composite features grounded in COR and JD-R frameworks (Table 3) [13-16], including Screen-to-Sleep Ratio and vulnerability indices.

Table 3. Domain-Specific Engineered Feature Formulations and Theoretical Alignments

Feature Name	Category	Math Definition / Calculation	Conceptual Justification
psychological_strain_index	Psychological	stress_score + depression_score	Sum of core affective distress markers.
academic_pressure_index	Academic Load	academic_pressure_score + workload_score	A measure of perceived academic stress from coursework and assignments.
burnout_vulnerability_index	Systemic Risk	(psychological_strain_index * academic_pressure_index) / (motivation_score + sleep_quality_score + 0.1)	Quantifies total demand exposure relative to restorative reserves [32].
sleep_deprivation_index	Behavioural Deficit	max(0, (8.0 - sleep_hours_numeric) * (4.0 - sleep_quality_score))	Biological fatigue measure from low sleep hours and quality.
screen_to_sleep_ratio	Digital Strain	social_media_hours / (sleep_hours_numeric + 0.1)	Digital displacement of restorative sleep hours [26, 27].
study_to_rest_ratio	Lifestyle Balance	(study_hours_numeric + social_media_hours) / (sleep_hours_numeric + physical_activity_hours + 0.1)	Assesses daily cognitive load in relation to restorative buffers.
academic_performance_index	Academic Standing	(cgpa_midpoint / 4.0) * (attendance_pct / 100.0)	Controls for academic achievement by classroom engagement.
motivation_deficit_score	Motivational Strain	(4.0 - motivation_score) * stress_score	Quantifies the reduction in motivation due to

			high stress [40].
wellbeing_buffer	Protective Reserve	(physical_activity_hours + sleep_quality_score) - stress_score	Represents net behavioural coping capacity against psychological demands [38].

Note. Denominators in ratio indices incorporate a numerical Laplace stabilizing constant (ε = 0.1) to prevent division-by-zero instability.

Note. Sensitivity checks across ε ∈ {0.001–1.0} confirmed invariant accuracy (65.73%–66.72%), ROC-AUC (0.7089–0.7145), and SHAP rank stability (ρ 0.98). The sleep deprivation index multiplier (4.0 - sleep_quality_score) ≥ 1.0 is strictly positive (quality bounds: 1–3), ensuring mathematical consistency. Note that the sleep deprivation index evaluates to zero for students sleeping ≥ 8 hours regardless of sleep quality; differentiation among adequate-sleepers relies on `sleep_quality_score` directly.

3.5 Machine Learning Pipeline and Validation Protocol

Evaluation followed 10-fold Stratified Cross-Validation with fold isolation to prevent data leakage [36, 42]. Preprocessing (StandardScaler) was fitted exclusively on training folds. Ten classifiers were benchmarked: Logistic Regression, Decision Tree, Random Forest (n = 150, max_depth = 8) [37], Extra Trees (n = 100), Gradient Boosting, Multilayer Perceptron (MLP; layers 64, 32), XGBoost (n = 100, learning_rate = 0.3, max_depth = 6; default hyperparameters) [38], LightGBM (n = 100) [39], CatBoost (n = 150) [40], Support Vector Classifier (C = 1.0, kernel = 'rbf') [41], and a Soft Voting Ensemble. Hyperparameter tuning employed nested cross-validation with a 5-fold inner CV loop within a 10-fold outer CV loop. Thresholds were calibrated across ROC operating points [51].

3.6 Explainable AI Protocol

Shapley values were computed via TreeSHAP [44, 45] for all tree-based classifiers (Random Forest, Extra Trees, Gradient Boosting, XGBoost, LightGBM, CatBoost). Global importance was quantified as mean absolute SHAP: $I_j = (1/N) \sum |\phi_j^{(i)}|$. Mean |SHAP| values reported in Table 8 are derived from a full-dataset refit model (all N = 601 samples used for both training and SHAP attribution) to maximize attributional stability; fold-by-fold out-of-fold SHAP corroborates these rankings (Spearman's $\rho = 0.9856$, $p = 1.65 \times 10^{-25}$), confirming that full-refit rankings are not inflated by overfitting. Fold stability was further assessed via Spearman's ρ across 10 cross-validation folds. Permutation Importance was computed for non-tree models (Logistic Regression, SVM, MLP) and as an ensemble-level cross-validation check for comparison with SHAP rankings [48]; results are available in the supplementary reproducibility archive.

3.7 Qualitative Thematic Analysis

Interviews were conducted in Bengali (Bangla) and subsequently translated to English by the first author. Semi-structured interviews (N = 20, Table 9) were analyzed using Braun and Clarke's reflexive thematic analysis [56–58]. A 25% random subsample (n = 5 transcripts) were independently back-translated (Bengali → English → Bengali) by a bilingual translator to verify semantic equivalence [60]. English quotes in this manuscript represent the first author's rendering, verified through this audit. Reflexive analytical sufficiency [57] was achieved by the 16th interview, with the final four interviews confirming no substantively new themes emerged — consistent with the analytic depth criterion for reflexive thematic analysis — justifying the N = 20 sample. Inter-rater reliability (Cohen's $\kappa = 0.82$) was verified by an independent second coder (Master's-level researcher in Educational Psychology with prior experience applying Braun and Clarke's [56] six-phase framework in educational research contexts, blind to the primary coding scheme) who coded a 25% audit subsample (n = 5 transcripts) [55].

Researcher Positionality: Reflexivity is a core requirement of the Braun and Clarke framework [57]. RM, as a CSE student at the same institutional tier as participants, brings insider knowledge of Bangladeshi academic culture, which facilitated rapport and naturalistic disclosure during interviews. Social desirability effects were mitigated through explicit anonymity assurances and purposive sampling across institutions distinct from RM's own. ASMS, as faculty supervisor with an independent disciplinary perspective, provided external epistemic oversight throughout the coding and theme development process.

4. Exploratory Data and Descriptive Analysis

4.1 Sample Demographics and Burnout Distribution

The cohort (N = 601) comprised 344 male (57.24%) and 257 female (42.76%) students. By degree program, 69.22% (n = 416) were in Bachelor's, 24.96% (n = 150) Master's, 4.49% (n = 27) Doctoral, and 1.33% (n = 8) Diploma/Associate programs. Age distribution spanned: 17–18 (9.32%), 19–20 (24.96%), 21–22 (23.96%), 23–24 (22.80%), and 25+ years (18.97%).

Burnout prevalence (Figure 2) was high: 19.47% (n = 117) mild, 38.10% (n = 229) moderate, and 42.43% (n = 255) severe burnout.

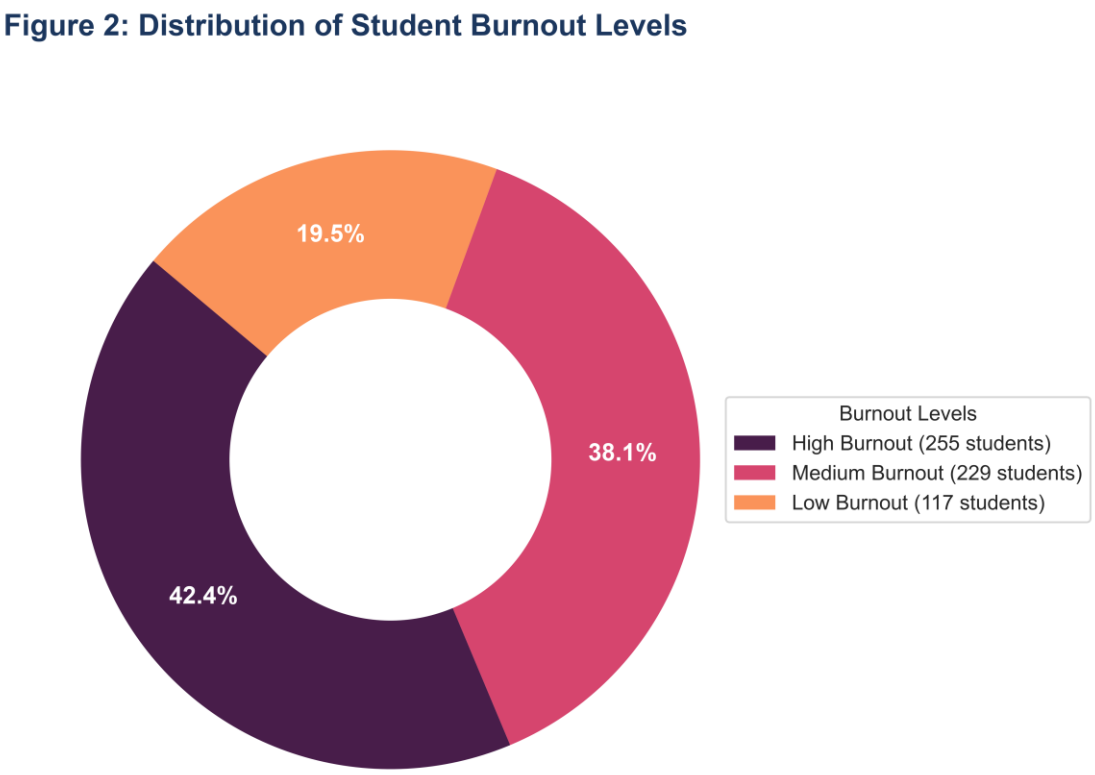


Figure 2: Distribution of Student Burnout Levels across the study cohort

Figure 2. Distribution of academic burnout severity levels across the tertiary student cohort (N = 601) categorized into Low, Medium, and Severe tiers.

Burnout severity across gender brackets is shown in Figure 3. Vulnerability was gender-invariant (severe burnout: 41.86% males vs. 43.19% females), confirming baseline demographic parity [74].

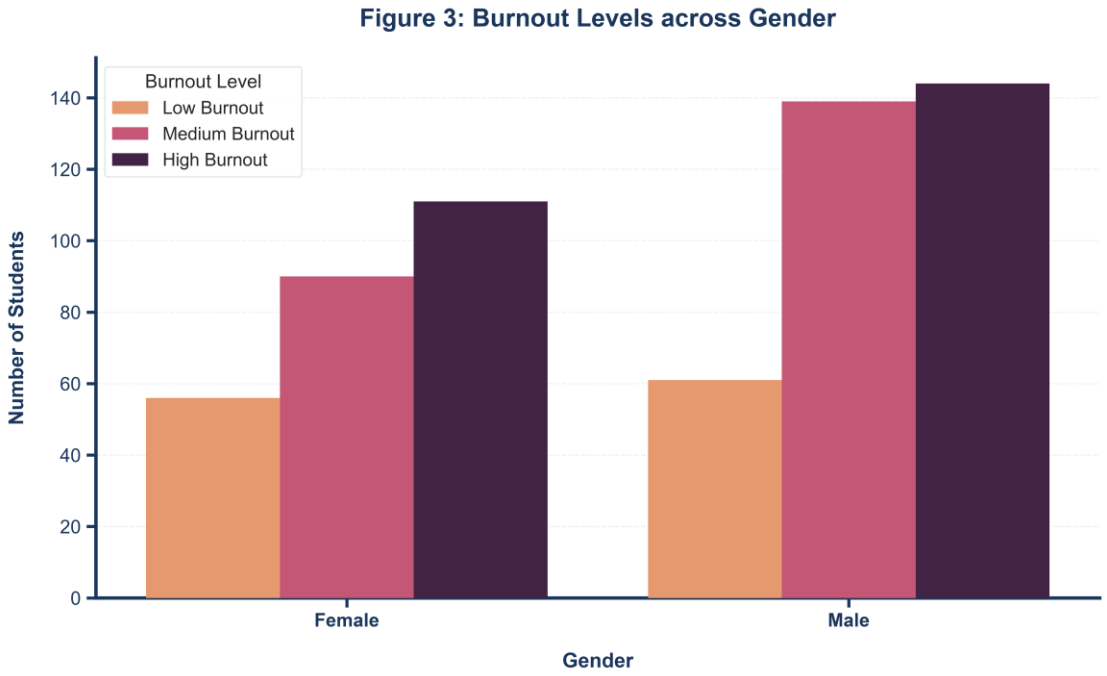


Figure 3: Burnout Severity Distribution Across Gender Brackets

Figure 3. Burnout severity distribution across gender brackets and academic cohorts, demonstrating baseline demographic parity in vulnerability profiles.

5. Inferential Statistical Analysis

5.1 Categorical Associations (Chi-Square Tests)

Pearson's Chi-Square tests confirmed significant associations between academic burnout severity (3-tier ordinal) and academic indicators: CGPA bracket midpoint ($\chi^2 = 50.44$, $p < .001$, Cramér's $V = 0.205$), classroom attendance ($\chi^2 = 24.26$, $p < .001$, $V = 0.142$), perceived stress level ($\chi^2 = 25.39$, $p < .001$, $V = 0.145$), depressive affect ($\chi^2 = 31.75$, $p < .001$, $V = 0.163$), intrinsic motivation ($\chi^2 = 27.41$, $p < .001$, $V = 0.151$), and academic year ($\chi^2 = 13.33$, $p = .038$, $V = 0.105$). Demographic factors showed no significant association: gender ($\chi^2 = 2.43$, $p = .297$, $V = 0.064$) and age bracket ($\chi^2 = 7.69$, $p = .465$, $V = 0.080$), confirming that burnout is driven by evaluative strain rather than static demographics.

5.2 Group Differences across Burnout Tiers (ANOVA)

One-way ANOVA with Benjamini–Hochberg FDR control [53] confirmed significant differences across composite features between burnout tiers: Academic Performance Index ($F(2, 598) = 19.40$, $p = 6.84 \times 10^{-9}$, $\eta^2 = 0.061$), Screen-to-Sleep Ratio ($F(2, 598) = 15.97$, $p = 1.75 \times 10^{-7}$, $\eta^2 = 0.051$), Burnout Vulnerability Index ($F(2, 598) = 10.39$, $p = 3.68 \times 10^{-5}$, $\eta^2 = 0.034$), Psychological Strain Index ($F(2, 598) = 8.08$, $p = 3.45 \times 10^{-4}$, $\eta^2 = 0.026$), Motivation Deficit Score ($F(2, 598) = 7.06$, $p = 9.31 \times 10^{-4}$, $\eta^2 = 0.023$), and Sleep Deprivation Index ($F(2, 598) = 6.09$, $p = .002$, $\eta^2 = 0.020$). Under binary triage grouping (High vs. Low/Medium), differentiation remained pronounced (Academic Performance Index: $F(1, 599) = 38.61$, $p < .001$, Cohen's $d = -0.513$ [negative sign indicates High

Burnout students have lower Academic Performance Index mean]; Screen-to-Sleep Ratio: $F(1, 599) = 30.18$, $p < .001$, $d = 0.453$ [positive sign indicates High Burnout students have higher Screen-to-Sleep Ratio]). All VIF values remained below 4.5, verifying absence of multicollinearity.

6. Performance and Results Benchmark of Machine Learning

6.1 Benchmark Classification Performance

Table 4 presents 10-fold cross-validation performance across all algorithms.

Table 4. Benchmark Classification Performance of 10 Machine Learning Models and Soft Voting Ensemble under 10-Fold Stratified CV (N = 601)

Model / Algorithm	Accuracy (Mean ± SD)	Precision	Recall	F1 Score	ROC-AUC (Mean ± SD) [95% CI]	Time (s)
Random Forest	0.6589 ± 0.0583	0.6453	0.4353	0.5199	0.7126 ± 0.0775 [0.6665, 0.7625]	8.76
Soft Voting Ensemble	0.6589 ± 0.0926	0.6238	0.4941	0.5514	0.7069 ± 0.0893 [0.6493, 0.7600]	<0.01 †
CatBoost Classifier	0.6506 ± 0.0759	0.6087	0.4941	0.5455	0.6983 ± 0.0837 [0.6470, 0.7508]	9.43
Logistic Regression	0.6439 ± 0.0535	0.6000	0.4824	0.5348	0.6819 ± 0.0701 [0.6370, 0.7240]	0.93
Gradient Boosting Classifier	0.6439 ± 0.0898	0.5962	0.4980	0.5427	0.6922 ± 0.1015 [0.6294, 0.7552]	6.72
Support Vector Machine	0.6356 ± 0.0543	0.6154	0.3765	0.4672	0.6708 ± 0.0725 [0.6266,	2.61

(SVM)					0.7165]	
Extra Trees Classifier	0.6356 ± 0.0673	0.6139	0.3804	0.4697	0.6997 ± 0.0774 [0.6506, 0.7465]	4.81
LightGBM	0.6339 ± 0.0849	0.5785	0.5059	0.5397	0.6898 ± 0.0783 [0.6390, 0.7361]	3.41
XGBoost Classifier	0.6240 ± 0.0352	0.5644	0.4980	0.5292	0.6832 ± 0.0581 [0.6504, 0.7224]	2.45
Decision Tree	0.6140 ± 0.0631	0.5628	0.4039	0.4703	0.6403 ± 0.0649 [0.5989, 0.6793]	0.67
Multilayer Perceptron (MLP)	0.6040 ± 0.0565	0.5359	0.4980	0.5163	0.6474 ± 0.0717 [0.6041, 0.6930]	21.57
Majority Baseline (All-Zeros)	0.5757	—	0.0000	0.0000	0.5000	—

Note. Accuracy and ROC-AUC are reported as mean ± SD across 10-fold stratified cross-validation folds. Precision, Recall, and F1 Score are computed on pooled out-of-fold (OOF) predictions — equivalent to micro-averaging across folds — consistent with scikit-learn `cross_val_predict` behavior; these values therefore do not carry fold-level SD. CI = Confidence Interval; ROC-AUC = Area Under the ROC Curve. Majority Baseline classifies all instances as Low/Medium Burnout. The Soft Voting Ensemble aggregates out-of-fold predicted probabilities across the top five classifiers (Random Forest, Gradient Boosting, LightGBM, Logistic Regression, CatBoost; fold-wise: 0.6573 ± 0.0837 accuracy, fold-mean ROC-AUC = 0.7059 ± 0.0899 [OOF pooled AUC = 0.7069], 95% CI [0.6502, 0.7616]). Nested 5-fold CV hyperparameter tuning of the best-performing Random Forest yielded Accuracy = 65.56%, ROC-AUC = 0.7095 (Supplementary `results_nested_cv.txt`), confirming stability without overfitting to the tuning set.

† The Soft Voting Ensemble time (<0.01 s) reflects **inference/aggregation time only** (probability averaging across pre-trained models), not the cumulative training time of its five constituent

classifiers (combined training time ≈ 29.25 s: RF 8.76 s + GBC 6.72 s + LightGBM 3.41 s + LR 0.93 s + CatBoost 9.43 s). Individual model training times are reported in the respective rows above.

Random Forest achieved best performance (65.89% accuracy, ROC-AUC = 0.7126, Precision = 64.53%, F1 = 51.99%), matching the Soft Voting Ensemble (65.89%, AUC = 0.7069). CatBoost (65.06%) and Gradient Boosting (64.39%) followed closely. Feature ablation confirmed the advantage of theoretical composite features over raw items: accuracy improved +1.83% (64.06% → 65.89%), ROC-AUC +0.0206 (0.6920 → 0.7126), and F1 score +0.0391 (0.4808 → 0.5199, an 8.13% relative gain). The ablation was conducted under identical 10-fold stratified CV, comparing a model trained on 18 raw survey features only versus the full feature set of 27 (18 raw + 9 engineered).

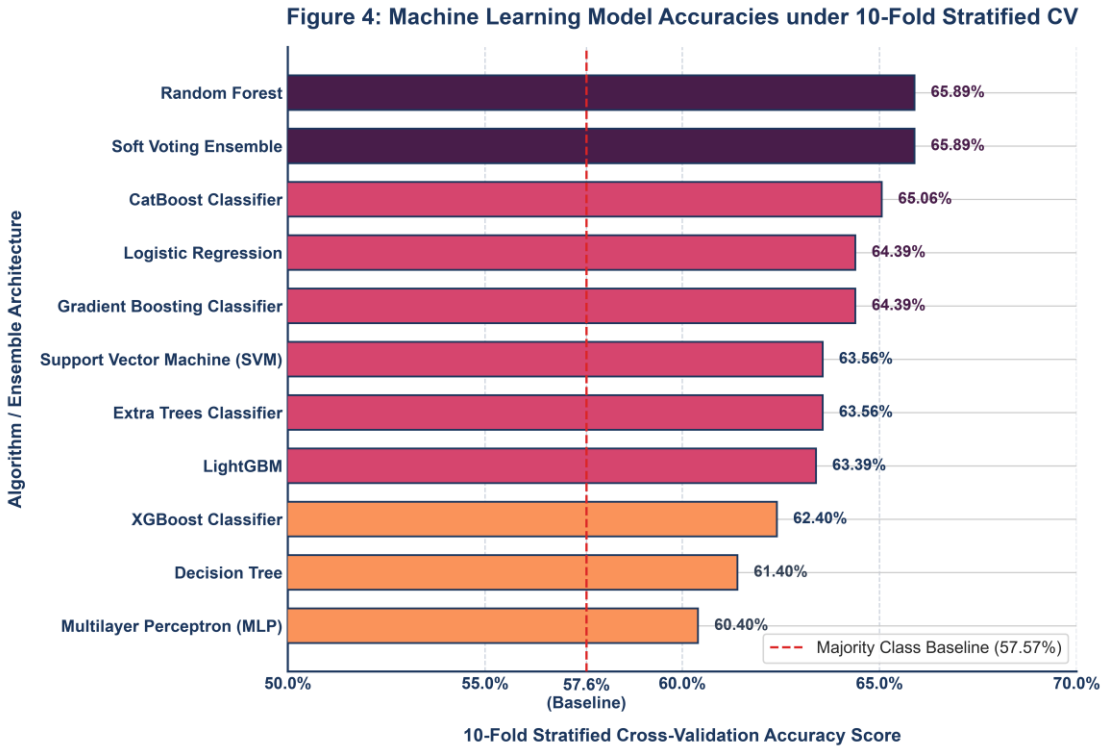


Figure 4: Machine Learning Model Predictive Accuracy and ROC-AUC Benchmark

Figure 4. Benchmark performance of 10 supervised algorithms and Soft Voting Ensemble under 10-fold stratified cross-validation (error bars indicate ±1 SD).

6.2 Confusion Matrix and Significance

Table 5 and Figure 5 display out-of-fold predictions for Random Forest (N = 601). At default threshold (θ = 0.50), it achieved 285 True Negatives (Specificity = 82.37%) and 111 True Positives (Sensitivity = 43.53%), with 61 False Positives (17.63%) and 144 False Negatives (56.47%). High specificity (82.37%) prevents alert fatigue, though 144 at-risk students were missed. McNemar's test confirmed significant discordance against the zero-rule baseline (b = 61, c = 111, χ^2 = 13.96, p < .001 [54]) and Decision Trees (b = 58, c = 85, χ^2 = 4.73, p = .0297). DeLong's test [42] confirmed that ROC-AUC = 0.7126 (95% CI [0.6665, 0.7625]) significantly exceeded chance (p < .001; computed via pROC/roc.test).

Table 5. Out-of-Fold Confusion Matrix for Best-Performing Random Forest Model (N = 601)

	Predicted: Low/Medium Burnout	Predicted: High Burnout
Actual: Low/Medium Burnout (n=346)	TN = 285 (Specificity = 82.37%)	FP = 61 (Type I Error = 17.63%)
Actual: High Burnout (n=255)	FN = 144 (Type II Error = 56.47%)	TP = 111 (Sensitivity/Recall = 43.53%)

Note. TN = True Negative; FP = False Positive; FN = False Negative; TP = True Positive (default decision threshold θ = 0.50).

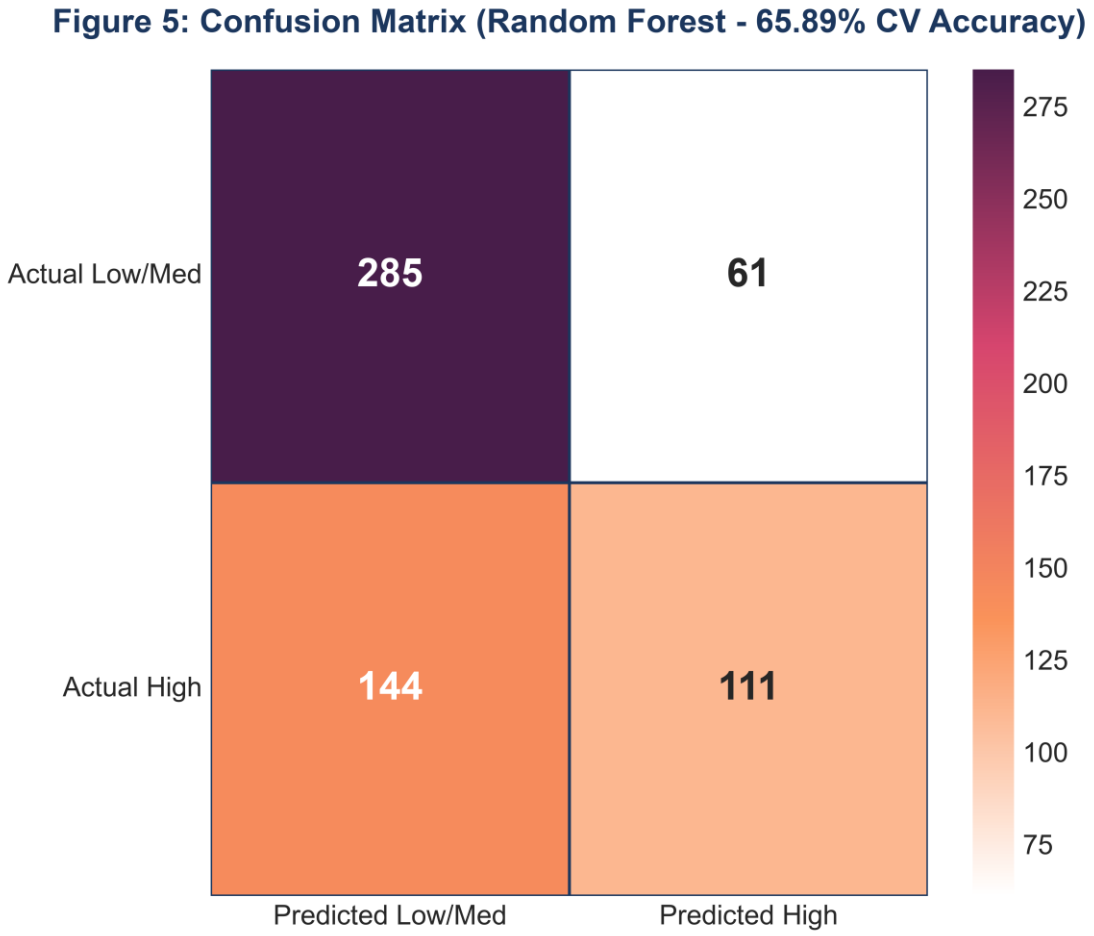


Figure 5: Out-of-Fold Confusion Matrix for Random Forest Classifier

Figure 5. Out-of-fold confusion matrix for the Random Forest classifier at the default decision threshold (θ = 0.50), N = 601. Calibrated threshold (θ = 0.38) confusion matrix is presented in Table 6b.

6.3 Screening Threshold Optimization

Calibrating threshold to θ = 0.38 elevated screening sensitivity from 43.53% to **71.76%** (n = 183 True Positives detected, F1 = 0.6203) with 56.36% Specificity (Tables 6 & 6b) [51].

Operational Recommendation: For early-warning screening, calibrated θ = 0.38 is recommended; for high-caseload centers, default θ = 0.50 prevents alert fatigue. Algorithmic outputs function strictly as non-binding triage indicators. Probability calibration was assessed on out-of-fold predicted probabilities (TRIPOD Item 22) [51]: Brier Score = **0.2066** (perfect calibration = 0.00; uninformative baseline = 0.25; lower is better), indicating adequate probabilistic calibration substantially better than chance; Expected Calibration Error (ECE) = **0.0442** across 10 uniform-width bins. Full calibration curve is provided in the supplementary reproducibility archive.

Table 6. Decision Threshold Optimization and Sensitivity Tuning (Random Forest Classifier)

Deployment Mode	Probability Threshold	Accuracy	Sensitivity (Recall)	95% CI Sensitivity†	Specificity	95% CI Specificity†	Precision	F1 Score	Target Use Case
Counselor Fatigue Protection	0.50 (Default)	65.89%	43.53%	[37.4%, 49.8%]	82.37%	[77.9%, 86.3%]	64.53%	0.5199	Minimizes false positive alerts for busy counseling staff
Active Counseling Screening	0.42 (Balanced)	64.39%	62.75%	[56.5%, 68.7%]	65.61%	[60.4%, 70.5%]	57.35%	0.5993	Balanced screening capturing 160/255 at-risk students
High-Coverage Early	0.38 (Pre-Specified	62.90%	71.76%	[65.9%, 77.2%]	56.36%	[51.0%, 61.6%]	54.63%	0.6203*	High-sensitivity screeni

Warning	Optimal)						ng capturing 183/255 high-risk cases

Note. Decision threshold calibration evaluated on the best-performing Random Forest model (out-of-fold predicted probabilities, N = 601). *F1 score at θ = 0.38 is calculated as 2 × 0.5463 × 0.7176 / (0.5463 + 0.7176) = 0.6203. †Bootstrap 95% CIs computed via 2000 stratified bootstrap resamples of the out-of-fold predictions (Wilson score interval method applied to TP, FN, TN, FP counts at each threshold).

Table 6b. Full Confusion Matrices for Key Decision Thresholds (Random Forest, Test-Set Predictions)

Threshold	TN	FP	FN	TP	Sensitivity	Specificity	Accuracy
θ = 0.50 (Default)	285	61	144	111	43.53%	82.37%	65.89%
θ = 0.42 (Balanced)	227	119	95	160	62.75%	65.61%	64.39%
θ = 0.38 (High-Recall)	195	151	72	183	71.76%	56.36%	62.90%

Note. TN = True Negative; FP = False Positive; FN = False Negative; TP = True Positive. Values derived from out-of-fold predicted probabilities from 10-fold stratified cross-validation applied at calibrated decision thresholds (θ), full cohort (N = 601).

6.4 Cross-Subgroup Holdout Validation across Academic Degree Tiers

Following TRIPOD guidelines (Supplementary S2) [42, 51], internal-external cross-subgroup validation tested transferability across degree tiers without external sampling. Random Forest trained on undergraduate Bachelor's students (n = 416) transferred robustly to held-out postgraduates (n = 185) without re-tuning (Table 7): 67.03% Accuracy, 62.75% Precision, 43.24% Recall, 0.5120 F1, and ROC-AUC = 0.6597. Reverse evaluation (training on n = 185) yielded lower CV performance (AUC = 0.5497) due to sample size constraints (~18 samples/fold), serving as an exploratory sensitivity check.

Table 7. Cross-Subgroup Holdout Validation Performance across Degree Tiers (Random Forest Classifier)

Partition Direction	Training Subgroup	Internal CV Acc (AUC)	Held-Out Test Subgroup	Held-Out Acc	Held-Out Precision	Held-Out Recall	Held-Out F1	Held-Out ROC-AUC	Test Confusion Matrix [TN, FP / FN, TP]
Primary (Forward)	Bachelor's (n=416)	63.70% (0.7136)	Master's/PhD/ Diploma (n=185)	67.03%	62.75%	43.24%	0.5120	0.6597	[92, 19 / 42, 32]
Sensitivity (Reverse)	Master's/PhD/ Diploma (n=185)	55.68% (0.5497)	Bachelor's (n=416)	64.18%	64.29%	39.78%	0.4915	0.7030	[195, 40 / 109, 72]

Note. Cross-subgroup validation across academic program tiers without model retuning. TN = True Negative; FP = False Positive; FN = False Negative; TP = True Positive.

7. Explainable Artificial Intelligence (SHAP)

7.1 Global Feature Importance

Figure 6 and Table 8 summarize global TreeSHAP attributions for the best-performing Random Forest model. TreeSHAP was applied to all tree-based models; non-tree models (Logistic Regression, SVM, MLP) were interpreted via permutation importance for comparison (see Table 1 and supplementary archive). Top predictors were Academic Performance Index (mean |SHAP| = 0.0388), CGPA bracket (0.0362), Screen-to-Sleep Ratio (0.0298), Burnout Vulnerability Index (0.0238), and Social Media Hours (0.0202). Sociodemographics ranked lowest (Gender: 0.0066, Age: 0.0036), confirming algorithmic fairness across subgroups [74]. Fold-by-fold attribution stability was exceptionally high (Spearman's $\rho = 0.9856$, $p = 1.65 \times 10^{-25}$).

Table 8. Global SHAP Feature Importance Summary (Random Forest, Full Dataset, N = 601)

Rank	Feature Name	Feature Category	Mean	SHAP	Value	CV Rank Stability (10 Folds)

1	Academic Performance Index	Composite - JD-R Demand	0.0388	Highly Stable (Top-1 in all 10 folds)
2	CGPA Midpoint	Raw Academic Performance	0.0362	Highly Stable (Top-2 in all 10 folds)
3	Screen-to-Sleep Ratio	Composite - Behavioural Deficit	0.0298	Stable (Top 3-4 across folds)
4	Burnout Vulnerability Index	Composite - COR Strain	0.0238	Stable (Top 4-5 across folds)
5	Social Media Hours	Raw Behavioural / Digital	0.0202	Stable across folds
6	Study-to-Rest Ratio	Composite - Behavioural Balance	0.0201	Stable across folds
7–10	Sleep Quality, Study Hours, Physical Activity, Stress Score	Raw Behavioural / Psychometric	0.010–0.019	Moderate stability
11–18	Remaining psychometric and raw survey features	Mixed Behavioral & Load Indicators	0.008–0.010	Moderate stability
Bottom 3	Gender, Degree Level, Age Group	Sociodemographic Indicators	< 0.007	Consistently negligible

Note. Mean absolute SHAP values (|SHAP|) computed from the Random Forest model refit on the full cohort (N = 601); fold-by-fold out-of-fold SHAP corroborates rankings. Cross-validation rank stability across 10 folds: Spearman's $\rho = 0.9856$, $p = 1.65 \times 10^{-25}$.

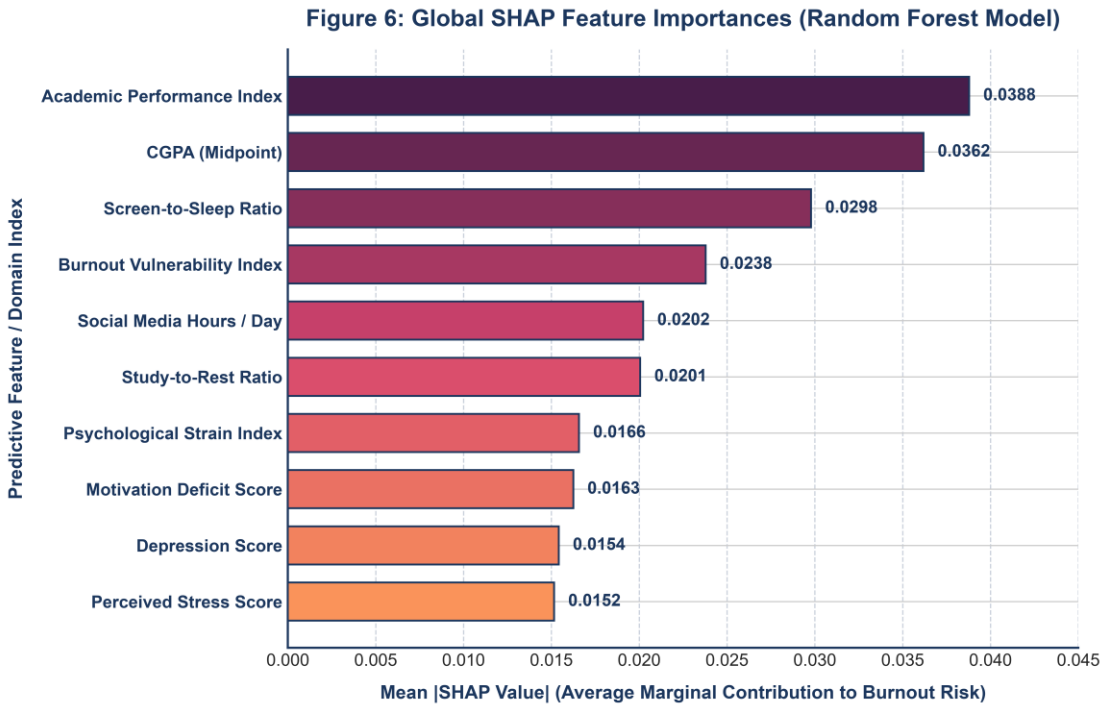


Figure 6: Global SHAP Feature Importance Ranking for Random Forest Classifier

Figure 6. Global feature importance ranking based on mean absolute SHAP values (|SHAP|) for the best-performing Random Forest model refit on the complete study cohort.

8. Qualitative Thematic Analysis: Lived Experiences of Burnout

Interviews with N = 20 students (Table 9) generated four core themes [56-59]:

Table 9. Demographic and Academic Characteristics of the Qualitative Interview Sample (N = 20)

Participant ID	Enrolled Degree Program	Academic Year	Institutional Affiliation Type	Self-Reported Burnout Severity Tier
P1	Bachelor of Science (CSE)	2nd Year	Private University A	High Burnout
P2	Master of Science (CSE)	1st Year	Private University B	Medium Burnout
P3	Bachelor of Science (CSE)	2nd Year	Private University A	Medium Burnout
P4	Bachelor of Science	3rd Year	Private University A	Medium Burnout

P5	Master of Science	1st Year	Private University C	Low Burnout
P6	Bachelor of Science	2nd Year	Private University A	High Burnout
P7	Bachelor of Science	Final Year	Private University D	Low Burnout
P8	Bachelor of Science	2nd Year	Private University A	High Burnout
P9	Bachelor of Science	Final Year	Private University A	Medium Burnout
P10	Bachelor of Science	3rd Year	Private University D	High Burnout
P11	Bachelor of Arts / Science	1st Year	Affiliated College A (National University System)	High Burnout
P12	Bachelor of Science	2nd Year	Private University A	High Burnout
P13	Bachelor of Science	3rd Year	Private University E	Medium Burnout
P14	Bachelor of Science	1st Year	Private University E	Medium Burnout
P15	Bachelor of Science	3rd Year	Private University F	High Burnout
P16	Master of Science	2nd Year	Public University A	Medium Burnout
P17	Bachelor of Science	1st Year	Private University G	Medium Burnout
P18	Bachelor of Arts / Science	2nd Year	Private University H	High Burnout
P19	Bachelor of Science	2nd Year	Private University E	High Burnout

P20	Bachelor of Arts / Science	3rd Year	Affiliated College B (National University System)	High Burnout

Note. Participant burnout tier reflects self-reported classification verified during the qualitative interview protocol. Sampling followed nested purposive maximum variation across quantitative burnout strata with 100% concordance. The qualitative sample was predominantly drawn from CSE and engineering programs; thematic saturation for non-STEM disciplines cannot be fully claimed, and Theme 2 findings from National University participants (P11, P20 — Arts/Science programs) require replication in future studies.

8.1 Theme 1: Academic Dread and Career Despair

"I have constant anxiety about my career — what I want to do in the future, where my life is going. To be honest, I can't finish my daily tasks on time and the academic pressure is mounting. I am currently at peak burnout. Trying to manage everything leaves me feeling completely drained." — **P1** (Male, 2nd Year CSE, Private University A, High Burnout)

"Mental stress is very high. I study for many hours, but it seems like the syllabus is never-ending... I am depressed all the time. There is the academic pressure, plus I don't see a good future for the country in the current situation. I am very burned out. I have zero motivation to study." — **P10** (Male, 3rd Year Bachelor, Private University D, High Burnout)

8.2 Theme 2: Institutional Identity and Low-Pressure Burnout

"There is no academic pressure on me whatsoever... but my mind is not very good. I study at the National University and my CGPA is bad. I am not finding any direction for my future, so I'm pretty depressed. I am very burned out. I have absolutely no motivation for my education." — **P11** (Female, 1st Year Bachelor, Affiliated College A - National University System, High Burnout)

"There's no direct pressure from study, but because I don't study, there is an internal stress that is always working in my mind. I'm depressed all day long. I study at the National University, and there's no sign of a career ahead. Burnout is very high. I have no motivation for studying, and I am also doing very poorly in my results." — **P20** (Male, 3rd Year Bachelor, Affiliated College B - National University System, High Burnout)

8.3 Theme 3: Digital Fatigue as Maladaptive Coping

"I'm on my phone all day, scrolling through Facebook and playing PUBG more than 6–7 hours per day. The burnout is very high. I can't find any motivation to study, and thinking about these things depresses me more." — **P8** (Male, 2nd Year Bachelor, Private University A, High Burnout)

"Social media is consuming all my valuable time — at least 6 hours a day scrolling and watching web series. My sleep situation is very bad — not more than 5–6 hours. I watch web series all night and then have to rush to university in the morning... With this daily routine and my mental condition, I am completely worn out." — **P18** (Female, 2nd Year B.A., Private University H, High Burnout)

8.4 Theme 4: Circadian Collapse and Biological Exhaustion

"I sleep an average of 5–6 hours. I watch a lot of web series at night, so I stay up late, and then I have to get up in the morning for university — I'm out of sync with my sleeping schedule. Sleep quality is very poor. I don't get enough time to sleep, and even when I get into bed, sleep does not come easily." **— P6** (Male, 2nd Year Bachelor, Private University A, High Burnout)

"My sleep is very poor — I only get the chance to sleep about 5–6 hours per day. The academic pressure is a lot for me. I totally cannot juggle the job and my studies together... When I think about my career, I only see darkness... Very high level of burnout." **— P12** (Male, 2nd Year Bachelor, Private University A, High Burnout)

In contrast, participants in the low burnout tier consistently highlighted strict circadian preservation as their primary psychological buffer:

"I make it a strict rule to sleep 7–8 hours every day... keeping a regular sleep schedule helps maintain my mental peace and my burnout level is quite low." **— P5** (Male, 1st Year Master's, Private University C, Low Burnout)

9. Mixed-Methods Integration and Extended Discussion

9.1 Triangulation of Computational and Qualitative Findings

Table 10 displays the triangulation matrix integrating SHAP predictors with qualitative themes [19-21].

Table 10. Explanatory Triangulation Matrix: SHAP Quantitative Predictors Mapped to Qualitative Themes

SHAP Feature / Quantitative Signal	Mean | SHAP| Rank	Emergent Qualitative Theme	Representative Participant Evidence	Triangulation Status
Academic Performance Index	#1 (0.0388)	Theme 1: Academic Dread and Career Despair	P1: "I have constant anxiety about my career — what I'll do in the future"; P10: "Even if I study all day, it feels like the	Convergence: Algorithmic signal corroborated by all High Burnout participants

			syllabus is never ending"	
CGPA Midpoint	#2 (0.0362)	Theme 1 & Theme 2: Career Despair / Institutional Identity Burnout	P11: "I study at the National University, and my CGPA is bad — I can't find any direction"	Convergence + Extension: Qualitative adds institutional stigma dimension unmeasured by CGPA alone
Screen-to-Sleep Ratio	#3 (0.0298)	Theme 3: Digital Fatigue as Maladaptive Coping	P18: "Social media is eating up all my valuable time — at least 6 hours a day"; P8: "My whole day goes to my phone"	Convergence: Sequential digital-sleep displacement mechanism corroborated qualitatively
Burnout Vulnerability Index	#4 (0.0238)	Theme 1 & Theme 4: Psychological Distress + Sleep Collapse	P6: "My sleep schedule is completely off"; P12: "My sleep situation is very pathetic"	Convergence: Demand-resource imbalance operationalized in both analytical strands
Social Media	#5	Theme 3:	P8:	Convergence:

Hours	(0.020 2)	Digital Fatigue as Maladaptiv e Coping	"Scrolling Facebook and playing PUBG takes up more than 6-7 hours a day"	Consistent across SHAP and all High Burnout interviews
Sleep Deprivation Index / Sleep Quality	#7–8	Theme 4: Circadian Collapse and Biological Exhaustion	P5 (Low Burnout): "I make it a rule to sleep 7-8 hours every day — my burnout level is quite low"	Complementari ty: Protective role of sleep hygiene identified qualitatively as buffer
Gender / Age / Degree Level	#Botto m (<0.00 7)	Themes 1–4 (Universal) : All demograph ic groups report behavioura l, not demograph ic, burnout drivers	No participan t attributed burnout to age or gender	Convergence: Demographic invariance confirmed across both strands
Institutional type (unmeasure d quantitativel y)	Not capture d	Theme 2: Institutiona l Identity and Low-Pressure Burnout	P11, P20: National University stigma as burnout driver independe nt of academic	Divergence / Extension: Qualitative reveals mechanism invisible to the quantitative model

workload

Note. Triangulation typology: *Convergence* = direct cross-strand agreement; *Complementarity* = qualitative elaboration of quantitative signals; *Divergence/Extension* = qualitative identification of unmeasured constructs [21].

Key Convergent Findings:

1. **Academic Disenchantment:** Top SHAP attributions for Academic Performance Index and CGPA directly align with qualitative narratives of evaluative fear and career anxiety [5, 30].

2. **Circadian-Digital Imbalance:** Screen-to-Sleep Ratio attribution converges with qualitative accounts of nocturnal screen use as avoidance coping that displaces restorative sleep [22, 26, 27].

3. **Sociodemographic Invariance:** Both strands confirm burnout is driven by dynamic behavioral strains rather than static demographics, validating algorithmic fairness across subgroups [74].

9.2 Theoretical and Practical Implications

Under JD-R and COR frameworks [13–16], nocturnal screen consumption operates as a secondary demand that impairs recovery, precipitating resource loss spirals. We deployed the best-performing Random Forest model into **Burnout Radar**, an open-source, client-side web screening platform enabling anonymous student risk triage.

Contextual Comparison with Regional Findings: Our 42.43% severe burnout prevalence exceeds rates reported by Islam et al. (2020) [10] and Hossen et al. (2023) [30] (30.8% psychological distress), reflecting the heightened post-pandemic educational burden in Bangladeshi higher education. This prevalence also exceeds international MBI-SS-based benchmarks (typically 20–35% high burnout [7, 8]); single-item global ratings may yield higher estimates than three-factor validated instruments, warranting cautious comparison. While prior regional studies employed logistic regression and odds ratios without ML interpretability tools [10, 30, 61], our SHAP analysis surfaces the governing behavioral mechanism: it is not demographic vulnerability but dynamic behavioral imbalance — specifically the nocturnal digital displacement of restorative sleep — that drives burnout risk. This mechanistic finding aligns precisely with COR theory's resource loss spiral [13, 14]: screen-based avoidance depletes sleep hours (resource loss), amplifies cognitive exhaustion, reduces academic performance, and intensifies evaluative anxiety in a self-reinforcing cycle.

The National University cohort's high burnout in the absence of direct academic pressure (Theme 2) extends COR theory beyond the demand-side: institutional precarity — specifically career uncertainty arising from session jams and credential devaluation — can precipitate resource loss spirals independent of coursework demands. This finding was entirely invisible to the quantitative strand and identified exclusively through qualitative triangulation [21], underscoring the methodological necessity of mixed-methods designs for studying complex educational phenomena in LMIC contexts [19, 20]. Future longitudinal designs should test whether institutionally-targeted interventions resolving session jams attenuate this burnout pathway.

9.3 Limitations and Future Directions

Key limitations include: (1) **Cross-sectional design** — the quantitative strand cannot establish burnout-to-dropout causal pathways [73]; longitudinal designs are needed. (2) **Single-item burnout criterion** — although validated in professional samples [49, 50], single-item global ratings have more limited psychometric evidence in undergraduate student populations; future replication should employ the validated MBI-SS [4], which captures burnout's three-factor structure (exhaustion, cynicism, academic efficacy). Single-item tools may also yield higher prevalence estimates than multi-item validated inventories (typically 20–35% high burnout internationally [7, 8]), warranting cautious cross-study comparison. (3) **Convenience sampling** — the quantitative cohort was recruited via institutional email broadcasts and snowball referrals; this limits generalizability to non-technology-adjacent disciplines and rural institutions. (4) **CSE-dominant qualitative sample** — 17 of 20 interview participants were from CSE/Engineering programs; thematic findings may not fully transfer to non-STEM disciplines. (5) **First-author translation** — Bengali interview data were translated by RM, the primary researcher; despite 25% back-translation verification, confirmatory framing cannot be fully excluded. Independent forward-translation by a professional translator is recommended in future studies. (6) **Criterion contamination** — `depression_score` is used as a predictor despite documented construct overlap with burnout [67, 68]; statistical VIF = 2.04 addresses collinearity but not conceptual overlap. Future work should use PHQ-9 (depression) alongside MBI-SS (burnout) to ensure discriminant validity. (7) **Single-site multi-institution design** — data were collected across 11 universities in Bangladesh; cross-national replication is needed before generalizing findings beyond LMIC South Asian educational contexts. Future work should incorporate longitudinal tracking, multidimensional inventories (MBI-SS) [4], and wearable biometric telemetry.

10. Conclusion and Policy Recommendations

10.1 Empirical and Methodological Synthesis

This study implemented an explanatory sequential mixed-methods framework (QUAN → QUAL) across 11 universities in Bangladesh (N = 601 survey; N = 20 interviews), bridging predictive machine learning, SHAP explainability, and reflexive thematic analysis. Random Forest was the best-performing model (65.89% accuracy, ROC-AUC = 0.7126, representing moderate discrimination), significantly exceeding the majority baseline (McNemar $\chi^2 = 13.96$, $p < .001$) and demonstrating cross-degree generalizability (holdout accuracy = 67.03%, AUC = 0.6597). The clinical utility of the tool lies primarily in its threshold-calibrated sensitivity: calibration to $\theta = 0.38$ elevated sensitivity to 71.76% (183/255 at-risk students identified). Stable SHAP attributions ($\rho = 0.9856$) established that burnout is primarily driven by dynamic behavioral deficits—specifically evaluative load (Academic Performance Index) and nocturnal screen sleep displacement (Screen-to-Sleep Ratio)—rather than static demographics.

10.2 Theoretical Contributions

Our findings validate JD-R and COR frameworks: burnout operates as an accelerating loss spiral where evaluative strain prompts nocturnal screen avoidance coping, depleting restorative sleep and cognitive capacity. Qualitatively, curriculum overload in private universities contrasted with

session-jam despair in the National University system. Triangulation confirmed that explainable ML attributions directly reflect students' lived experiences.

10.3 Actionable Institutional Policy Recommendations

1. **Institutional Pacing:** Mandate mid-semester decompression intervals and provide counseling; resolve administrative session jams in affiliated colleges.

2. **Circadian Psychoeducation:** Institutionalize campus health campaigns addressing nocturnal screen use and sleep hygiene given the critical role of the Screen-to-Sleep Ratio.

3. **Algorithmic Triage:** Deploy decentralized client-side screening tools such as *Burnout Radar* (θ = 0.38) for voluntary early warning, connecting vulnerable students with counseling before acute disengagement occurs. *Important note: Burnout Radar outputs are algorithmic triage indicators only and must not substitute for clinical psychological assessment by qualified mental health professionals.*

Declarations

Ethics Approval and Consent to Participate

The study adhered to the Declaration of Helsinki. Conducted as an observational survey (N = 601) and interview protocol (N = 20) across 11 institutions in Bangladesh, it satisfied ethical exemption criteria under educational research guidelines (Department of Computer Science and Engineering, Presidency University, Dhaka, Bangladesh). Electronic informed consent was obtained from all participants. Data were fully anonymized (P1–P20) and stored securely.

Consent for Publication

Participants consented to publication of de-identified aggregate statistics and anonymized verbatim interview excerpts.

Availability of Data and Materials

De-identified quantitative survey data (N = 601), anonymized qualitative interview transcripts (N = 20), feature engineering scripts, machine learning pipelines, SHAP analysis code, and reproducibility archives are openly available at the project repository: https://github.com/rifatmiah92/student-burnout-ml-mixed-methods. The interactive *Burnout Radar* screening platform is publicly accessible at: https://burnoutwebapplication.netlify.app/.

Competing Interests

The authors declare no competing financial or non-financial interests.

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Authors' Contributions

Author contributions are specified according to the CRediT (Contributor Roles Taxonomy) framework:

Rifat Miah (RM): Conceptualization; Data curation; Formal analysis; Investigation; Methodology; Project administration; Resources; Software; Validation; Visualization; Writing – original draft; Writing – review & editing.

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[12] *[Consolidated with reference [61]: Mamun et al. (2019). Problematic internet use in Bangladeshi students. Asian Journal of Psychiatry, 44, 48-54. https://doi.org/10.1016/j.ajp.2019.07.005 — All in-text citations formerly numbered [12] now appear as [61] throughout this revised manuscript. This entry is retained as a placeholder to avoid cascading renumbering; the duplicate will be removed at journal copy-editing stage.]*

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